DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
MONDAY TEST

CLASS: X | **TOTAL MARKS: 40**

SUBJECT: ROBOTICS & AI | **DATE: 08/07/24**

Instructions: Attempt all questions. The intended marks for questions are given in brackets [ ]. Write Variable description table for each Python program. This paper consists of three printed pages.

Question 1 **[1x10=10]**

i) Which component is essential for a robot to move in a controlled manner?

a) Sensor b) Actuator
c) Processor d) Power Supply

ii) Which type of robot consists of multiple smaller robots working together to complete a task?

a) Humanoid Robot b) Teleported Robot
c) Mobile Robot d) Swarm Robot

iii) **Assertion (A):** Medical robots are increasingly used for performing surgeries.
Reason (R): Medical robots can perform surgeries without any human supervision.

a) Both A and R are true, and R is the correct explanation of A.
b) Both A and R are true, but R is not the correct explanation of A.
c) A is true, but R is false.
d) A is false, but R is true.

iv) What is the **Turing Test** used for?

a) To determine if a machine can exhibit intelligent behavior equivalent to, or indistinguishable from, that of a human
b) To measure the processing speed of a computer
c) To test the durability of a machine
d) To evaluate the memory capacity of a system

v) Which of the following is used to define a block of code in Python?

a) Curly braces b) Parenthesis
b) Brackets d) Indentation

vi) **Assertion (A):** AI systems can improve decision-making processes in critical areas such as healthcare.
Reason (R): AI systems can process large amounts of data quickly and identify patterns that humans might miss.
a) Both A and R are true, and R is the correct explanation of A.
b) Both A and R are true, but R is not the correct explanation of A.
c) A is true, but R is false.
d) A is false, but R is true.

vii) Identify the diagram of ultrasonic sensor from the given diagram-

a)


b)


c)


d)


viii) What is one of the primary ***ethical*** concerns related to the use of AI technology?
a) Decreased efficiency in tasks
b) Potential for bias and discrimination
c) Limited scalability
d) High cost of implementation

ix) Deep learning includes algorithms based on the concept of artificial neural network. (True or False)

x) The light sensor detects the presence of an object. (True or False)

Question 2

i) The following Python code prints a ***hollow square***. Some part of the code is missing. Fill in the blanks so that it meets the objective. After substitution study the code and answer the following questions-

```
size = 5
for i in range(size):
    for j in range(_?1?_):
        if i == ___?2?___ or i == size - 1 or j == ___?3?__ or j == size - 1:
            print("*", end=" ")
        else:
            print(" ", end=" ")
    print()
```

a) What is the expression/ value at ?1? [1]
b) What is the expression/ value at ?2? [1]
c) What is the expression/ value at ?3? [1]
d) What is the role of **end** in print statement? [1]
e) Draw the output. [1]

ii) The following program is checking if the given number is a **prime.** The code is erroneous, hence, not giving the right output. Correct the errors and re-write the code. Underline the corrections. [5]

```
define isPrime(n):
    if n < 1:
            return False
    for i in range(2, n):
            if n % i = 0:
            return False;
    return true
```

Question 3

i) Write a program in Python to generate **Fibonacci** series upto **'n'** terms. [7]
Input:
N= 8
Output:
1 1 2 3 5 8 13 21

ii) What is the purpose of **actuators** in robots? Name any two types of actuators. [3]

Question 4

i) Define a method in Python that checks if **'n'** is a **palindrome** number. *A palindrome number is a number that reads the same forward and backward.* Example 3, 5, 121, 23432 etc [7]

ii) Discuss some ***ethical*** issues in terms of AI. [3]